

e_{ml}★ — Minimal Anti-Holomorphic Extension of the EML Sheffer Operator

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GitHub: https://github.com/antparis/eml_star

Abstract

Odrzywołek (2026) showed that $\text{eml}(x, y) = e^x - \ln y$, together with the constant 1, generates all standard elementary functions via finite composition. We identify a structural limitation: eml is holomorphic, so complex conjugation $\bar{z}$, and real and imaginary parts are not reachable by finite eml -compositions. We introduce the companion operator $\text{eml}^\star(x, y) = e^x - \ln \bar{y}$, which acts as a mirror reflecting the imaginary axis. We prove: (i) $\bar{z} = 1 - \text{eml}^\star(0, \text{eml}(z, 1))$ at depth 2, conditional on $\text{Im}(z) \in [-\pi, \pi)$; (ii) $\{\text{eml}, \text{eml}^\star, 1\}$ is dense in $C(K, \mathbb{C})$ for every compact $K \subset \{z : \text{Im}(z) \in [-\pi, \pi)\}$ by Stone–Weierstrass; (iii) the exact branch limitation is $\text{Im}(z) \in [-\pi, \pi)$. A direct numerical experiment confirms Theorem 3.1 to machine precision: $\text{eml}^\star$ achieves $\text{MSE} = 5.89 \times 10^{-33}$ vs. 12.97 for eml alone — a ratio of 2.2×10^{33} . Theorems 2.1, 4.3, and Corollary 2.2 are unconditional.

1. Introduction

Odrzywołek [1] demonstrated that $\text{eml}(x, y) = e^x - \ln y$, together with the constant 1, is a continuous Sheffer operator: every standard elementary function is a finite composition of eml -nodes. The grammar is $S \rightarrow 1 \mid \text{eml}(S, S)$. This is the continuous analogue of the NAND gate in Boolean logic.

A natural question arises: does $\{\text{eml}, 1\}$ reach all continuous functions on $\mathbb{C}$, or only the holomorphic ones? The answer is: only the holomorphic ones. This note identifies the obstruction precisely and proposes a minimal fix — the companion operator $\text{eml}^\star$ — that closes the gap.

2. The Holomorphic Barrier

Let $E_0 = \{1\}$, $E_{n+1} = E_n \cup \{\text{eml}(f, g) : f, g \in E_n\}$, and $E = \bigcup_n E_n$.

Theorem 2.1 (Holomorphic closure). *Every element of E is holomorphic on its domain.*

Proof. The constant 1 is holomorphic. The maps $z \mapsto e^z$ and $z \mapsto \ln z$ (principal branch) are holomorphic. Holomorphic functions are closed under subtraction and composition; the claim follows by induction on depth. ■

Corollary 2.2. *The operators Re , Im , and $z \mapsto \bar{z}$ are not in E .*

Proof. These maps are non-holomorphic ($\partial_{\bar{z}} \{\bar{z}\} \text{Im}(z) = 1/2 \neq 0$), so no element of E equals them by Theorem 2.1. ■

Remark 2.3. Corollary 2.2 does not imply $\pi \notin E$. Odrzywołek shows $\pi \in E$ at depth $K > 53$ (Table 4 of [1]). The distinction is between $\text{Im}(\cdot)$ as an operator and π as a fixed real constant.

3. The Companion Operator $\text{eml}\star$

Motivation. eml cannot distinguish z from $\bar{z}$ — it only sees the holomorphic world. To recover the conjugate, one modification suffices: replace $\text{Log}(y)$ by $\text{Log}(\bar{y})$ in the second argument.

Definition. $\text{eml}\star(x, y) := e^x - \text{Log}(\bar{y})$, where Log denotes the principal logarithm ($\text{Arg} \in (-\pi, \pi]$).

Theorem 3.1 (Conjugate recovery at depth 2). *For all $z \in \mathbb{C}$ with $\text{Im}(z) \in [-\pi, \pi]$:*

$$\bar{z} = 1 - \text{eml}\star(0, \text{eml}(z, 1))$$

Proof. $\text{eml}(z, 1) = e^z$. Then $\text{eml}\star(0, e^z) = 1 - \ln(\text{conj}(e^z)) = 1 - \bar{z}$, provided $\text{Im}(\bar{z}) = -\text{Im}(z) \in (-\pi, \pi]$, i.e. $\text{Im}(z) \in [-\pi, \pi]$. ■

Theorem 3.2 (Exact safe domain). The identity of Theorem 3.1 holds if and only if $\text{Im}(z) \in [-\pi, \pi]$. At $\text{Im}(z) = \pi$ the principal Log produces a jump of $2\pi i$. The boundary $\text{Im}(z) = -\pi$ holds; $\text{Im}(z) = +\pi$ fails. Outside the strip, jumps of exactly $2\pi i$ occur.

Corollary 3.3. $|z|$ is reachable at depth 4 via $|z| = \sqrt[4]{z \cdot \bar{z}}$, where $\bar{z}$ comes from Theorem 3.1 and multiplication from the eml arithmetic of [1].

Depth table

Expression	Rel. depth*	Via
$\bar{z} = 1 - \text{eml}\star(0, \text{eml}(z, 1))$	2	Theorem 3.1
$\text{Re}(z) = (z + \bar{z})/2$	3	Corollary 3.3
$ z ^2 = z \cdot \bar{z}$	3	eml arithmetic + Thm 3.1
$ z = \sqrt[4]{z \cdot \bar{z}}$	4	sqrt via eml [1]

Relative depth counts $\text{eml}\star$ nodes only.

4. Topological Completeness via Stone–Weierstrass

Lemma 4.1. *Every rational number lies in E .*

Proof. $0 = \text{eml}(0, 1)$. 1 is given. Arithmetic follows from [1]. ■

Lemma 4.2. *Every polynomial in $\mathbb{Q}[z, \bar{z}]$ lies in the algebra generated by $\{\text{eml}, \text{eml}\star, 1\}$.*

Proof. $z \in E$. $\bar{z}$ by Theorem 3.1. Products and sums by eml arithmetic. ■

Theorem 4.3 (Stone–Weierstrass density). *For every compact $K \subset \mathbb{C}$ with $K \subset \{z : \text{Im}(z) \in [-\pi, \pi]\}$, the algebra generated by $\{\text{eml}, \text{eml}\star, 1\}$ is dense in $C(K, \mathbb{C})$.*

Proof. On K , Theorem 3.1 gives $\bar{z}$ exactly, so $A = \mathbb{Q}[z, \bar{z}] \subseteq \{\text{eml}, \text{eml}\star, 1\}$ pointwise. A satisfies Stone–Weierstrass [2]: (i) separates points; (ii) does not vanish; (iii) closed under conjugation. Hence A is dense in $C(K, \mathbb{C})$. ■

Remark 4.4. Theorem 4.3 does not contradict Theorem 2.1: the algebra is dense, but individual elements of E remain holomorphic. Density is an approximation result, not a representation result.

Remark 4.5 (Global extension). The restriction $\text{Im}(z) \in [-\pi, \pi]$ is essential. Density on arbitrary compacts in $\mathbb{C}$ would require tracking winding numbers continuously — impossible with finite compositions of analytic/anti-analytic operators (monodromy theorem). Open problem.

5. Numerical Experiment

We provide a direct machine-precision verification of Theorem 3.1 and contrast $\text{eml}\star$ with eml alone. The test is deterministic, parameter-free, and fully reproducible from the repository.

Setup. $N = 40$ points per axis, $\text{Re}(z) \in [-3, 3]$, $\text{Im}(z) \in [-\pi + 0.1, \pi - 0.1]$ (strictly inside the safe strip). Total: 1600 evaluation points. Target: $\bar{z}$.

Test 1 — with $\text{eml}\star$ (Theorem 3.1):

```
pred = 1 - eml_star(0, eml(z, 1))
```

Test 2 — with eml alone (Corollary 2.2):

```
pred = 1 - eml(0, eml(z, 1))
```

Results

Metric	$\text{eml}\star$ (with conjugate)	eml (without conjugate)
MSE on 1600 points	5.89×10^{-33}	12.97
Ratio $\text{eml} / \text{eml}\star$	—	2.2×10^{33}

Metric	eml★ (with conjugate)	eml (without conjugate)
Conclusion	Theorem 3.1 verified ✓	Corollary 2.2 confirmed ✓

Interpretation. The MSE of 5.89×10^{-33} with **eml★** is the floating-point floor — effectively zero. The error of 12.97 with **eml** alone is not a numerical instability: it is the fundamental impossibility established by Corollary 2.2. The ratio of 2.2×10^{33} is a computational proof that **eml** and **eml★** operate in categorically different function spaces on this target.

Reproducibility. Script: `eml_toolkit/direct_verification.py`. Requires NumPy only. Runtime < 1 second.

6. Summary

Result	Sec.	Status
Holomorphic closure of E	2	Unconditional theorem
$\text{Re}, \text{Im}, \text{conj} \notin E$	2	Unconditional corollary
$\bar{z} = 1 - \text{eml}\star(0, \text{eml}(z, 1))$ at depth 2	3	Conditional: $\text{Im}(z) \in [-\pi, \pi)$
Domain $\text{Im}(z) \in [-\pi, \pi)$ (half-open)	3	SymPy + interval verification
$ z $ reachable at depth 4	3	Unconditional corollary
Density of $\{\text{eml}, \text{eml}\star, 1\}$ in $C(K, \mathbb{C})$	4	Conditional on branch-safety lemma
Theorem 3.1 at machine precision (ratio 10^{33})	5	Numerical — fully reproducible

Open problems. (1) Does $E \cap \mathbb{R}$ coincide with the exp-log closure of $\{1\}$ over $\mathbb{Q}$? This is connected to the Schanuel conjecture. (2) A fully analytic interval-arithmetic proof of branch-safety for the deep addition and multiplication trees of [1] remains open.

Appendix A — Explicit Clean Arithmetic Blocks

The following constructions realize subtraction, negation, and $1/2$ as finite **eml**-trees. Branch safety verified at 100 decimal digits on 10^5 random points.

subtract(x, y) — Depth 4:

```
eml( eml(1, eml(eml(1,x), 1)), eml(y, 1) )
```

minus(x) — Depth 7:

```
eml( eml( eml(1,eml(eml(1,x),1)), eml(eml(1,eml(eml(1,1),1)),1) ),
1 )
```

half — Depth 6 (constant):

```
eml(1, eml(1, eml(1, eml(1, eml(1, eml(1, 1))))))
```

Caveat. These blocks close branch safety for effective depth ≤ 4 . Branch safety for the full addition ($K \approx 27$) and multiplication ($K \geq 17$) trees of [1] is verified numerically at 60 decimal places (zero violations, 1000 random points) but an analytic interval proof for arbitrary compacts remains open.

Acknowledgments

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References

- [1] A. Odrzywołek, *All elementary functions from a single operator*, arXiv:2603.21852v2 (April 2026).
 - [2] M. H. Stone, *The generalized Weierstrass approximation theorem*, Mathematics Magazine 21 (1948), 167–184.
 - [3] G. Terzo, *Some consequences of Schanuel’s conjecture in exponential rings*, Comm. Algebra 36 (2008), 1171–1189.
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Ancillary files: `branch_safety_final.py · verify_theorem4.py · eml_toolkit / direct_verification.py · eml_star_branch_safety_lemma_proof.pdf`
License: MIT · *GitHub:* github.com/antparis/eml_star

5b. Causal Confirmation (GP Experiment)

To provide experimental confirmation of Corollary 2.2, a causal GP experiment was conducted using the DEAP library. The setup used a single input z , with target $\text{conj}(Z)$ on the safe domain $\text{Im}(z)$ in $[-\pi + 0.1, \pi - 0.1]$. The primitives included the corrected complex versions of `eml` and `eml_star`. A total of 100 independent runs were performed (200 individuals, 30 generations).

In 90 out of 100 runs, the best tree contained `eml_star`. The median ATE (increase in MSE when replacing `eml_star` by `eml`) was **11.71**, with a 95% bootstrap confidence interval of **[9.64, 12.57]**. This result demonstrates that `eml_star` is structurally necessary to reach the anti-holomorphic target $\text{conj}(z)$. Replacing `eml_star` by `eml` produces a statistically significant degradation, confirming that the anti-holomorphic extension is not redundant but essential.

The experiment is fully reproducible from the ancillary notebook
CGIB_v8_corrected.ipynb.

7. An Operator for Branch Monitoring (eml^n)

Motivation. Although $\text{eml}^\star$ solves the holomorphicity barrier identified in Corollary 2.2, a major practical difficulty remains in Odrzywółek’s original framework: ensuring that deep expression trees (especially those implementing addition and multiplication) never cross the branch cut of the logarithm. To address this issue, we introduce a third operator specifically designed to monitor the argument of intermediate values.

Definition. $\text{eml}^n(x, y) := \exp(x) - i \cdot \text{Arg}(y)$, where Arg denotes the principal argument function with range $(-\pi, \pi]$.

This operator provides direct, decoupled access to the imaginary part of the logarithm. When combined with eml and $\text{eml}^\star$, it allows one to track the argument of any intermediate result during the evaluation of an EML expression tree.

Proposition 7.1 (numerical independence). The closest algebraic combination from $\{\text{eml}, \text{eml}^\star\}$ — namely $(\ln(\bar{z}) - \ln(z))/2$ — achieves $\text{MSE} = 1.30\text{e}+01$ on $i \cdot \text{Arg}(Z)$ over a 40×40 grid, while eml^n achieves $\text{MSE} = 0.00\text{e}+00$. A formal algebraic independence proof remains open.

Numerical evidence. On both rectangular grids inside the safe strip and full circles around the origin (including the branch cut), eml^n recovers $i \cdot \text{Arg}(z)$ to machine precision ($\text{MSE} = 0$), while combinations of eml and $\text{eml}^\star$ alone fail to do so.

Operator	MSE on $i \cdot \text{Arg}(Z)$
eml^n	0.00e+00
eml	1.33e+01
$\text{eml}^\star$	3.16e-01

Perspective. eml^n does not automatically prevent branch-cut violations. However, it provides the missing primitive needed to detect and potentially constrain the argument during evaluation. It can therefore serve as a diagnostic tool for future formal proofs of branch safety on deep EML expressions, and as a building block for more advanced monitoring or regularization techniques.

Reproducibility. Script: `eml_zero_verification_v1.ipynb`. Requires NumPy only.
Runtime < 1 second.

7. The Third Companion Operator eml^0

Motivation. The system $\{\text{eml}, \text{eml}^\star, 1\}$ provides holomorphic and anti-holomorphic coverage on the safe strip $\text{Im}(z) \in (-\pi, \pi)$. Both operators involve the full complex logarithm $\ln(y) = \log|y| + i \cdot \text{Arg}(y)$. Neither isolates the principal argument $\text{Arg}(y)$ alone.

Definition. $\text{eml}^0(x, y) := \exp(x) - i \cdot \text{Arg}(y)$, where Arg denotes the principal argument ($\text{Arg} \in (-\pi, \pi]$).

Theorem 7.1 (Independence). eml^0 is not constructible by finite composition from $\{\text{eml}, \text{eml}^\star, 1\}$.

Proof sketch. $\text{eml}^0(0, y) = 1 - i \cdot \text{Arg}(y)$. Since $\text{Arg}(y) = \text{Im}(\ln(y))$, and Im is not in E by Corollary 2.2, eml^0 cannot be reached by finite $\text{eml}/\text{eml}^\star$ compositions. ■

Theorem 7.2 (Polar decomposition). The system $\{\text{eml}, \text{eml}^\star, \text{eml}^0, 1\}$ gives access to $\exp(x)$, $\log|y|$, and $\text{Arg}(y)$ separately — the complete polar decomposition on the safe strip.

Numerical verification. On a 40×40 grid ($\text{Re}(z) \in [-3, 3]$, $\text{Im}(z) \in [-\pi+0.1, \pi-0.1]$) and on a full circle ($r \in [0.5, 2]$, $\theta \in [-\pi, \pi]$):

Operator	MSE on $i \cdot \text{Arg}(Z)$
eml^0	0.00e+00
eml	1.33e+01
$\text{eml}^\star$	3.16e-01

Open problem. Does a finite operator system cover all of $\mathbb{C}$ without monodromy restriction? The winding number is a discrete topological invariant — it appears structurally unreachable by finite compositions of analytic/anti-analytic operators (monodromy theorem). This remains an open problem.

Reproducibility. Script: `eml_zero_verification_v1.ipynb`. Requires NumPy only. Runtime < 1 second.

8. Analog Circuit Implementation of the EML Operator Family

The EML operator admits a direct analog realization. A single EML gate is assembled from an exponential module (BJT antilog), a natural-logarithm module (BJT log feedback), and an operational-amplifier differential subtractor. The gate requires two to three op-amps and one or two transistors.

Complex expressions are realized by connecting gates in a binary-tree topology: the output voltage of each gate feeds directly into the inputs of the next. No clock or time-division multiplexing is employed. With 10 MHz operational amplifiers the propagation delay is $n \times 50\text{--}200$ ns per stage, n being the tree depth.

The anti-holomorphic extension $\text{eml}^\star$ is obtained by routing parallel channels for the real and imaginary parts together with a 180 degree phase inverter on the imaginary channel to realize conjugation. Numerical simulation confirms correct behavior on 10 test inputs with zero branch violations.

A known limitation is dynamic range overflow in deep trees: exponential accumulation causes divergence for inputs with real part greater than 1 at depth 3. Inter-stage clipping or normalization is required in practice.

Operator	Components needed	Complexity	Domain
eml	2-3 op-amps, 1-2 transistors	Low	Real, positive
eml★	Parallel Re/Im channels + 180 degree inverter	Medium	Complex, $\text{Im}(z)$ in $(-\pi, \pi)$
eml0	Additional argument-extraction circuitry	High	Complex (open)

The principal-argument extraction required for a complete analog implementation of eml0 remains an open problem.

Reproducibility. Script: analog_eml_circuit.py. Requires NumPy only. Runtime < 1 second.